GAJANAN SADAPHAL

Embedded Systems Enthusiast | IoT & Automation

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PROFILE

Recently graduating ENTC engineer with a strong interest in Embedded Systems, IoT, and Automation, possessing hands-on experience with PIC, LPC, and STM32 microcontrollers and embedded tools. Seeking internship or entry-level opportunities in embedded domain.

EDUCATION

2022–2026 | Bachelor of Engineering – Electronics & Telecommunication| 8.88 CGPA

Jaihind College of Engineering , Pune (SPPU University)

2020–2022 | 12th-Science

R.B. Narayanrao Borawake College,Shrirampur

TECHNICAL SKILLS

- Microcontrollers: PIC, LPC, STM32, Arduino
- Embedded Programming: Embedded C, C, C++ , java
- Embedded IDEs & Tools: MPLAB (PIC), Keil (LPC), STM32CubeIDE
- Simulation & Design Tools: Proteus, Multisim Live
- Communication Protocols: UART, SPI, I²C
- Sensors & Modules: Temperature, Gas, IR, LDR, Pulse Sensor, LCD
- Domains: Embedded Systems, IoT, Automation

INTERN WORK EXPERIENCE

Embedded Systems Intern – Emertxe Information Technologies February 2024 – March 2024)

- Developed and tested firmware for microcontrollers using Embedded C.
- Designed basic circuit schematics and performed hardware interfacing.
- Debugged firmware and resolved hardware-related issues.
- Worked with microcontroller peripherals such as GPIO, timers, and ADC.
- Gained practical exposure to embedded development workflows.

Junior Embedded Engineer (Intern) – OmegaSoft Technologies(December 2024 – February 2024)

- Worked extensively on Arduino-based embedded systems.
- Developed a Smart Home Automation system using Arduino and sensors.
- Interfaced sensors, relays, and actuators to control household appliances.
- Implemented automation logic to enhance convenience, energy efficiency, and security.
- Tested, debugged, and validated system performance through hands-on implementation.

PROJECTS

- **Smart Home Energy Management System using IoT**
  - Developed an IoT-based system with DHT11 sensor to monitor real-time temperature, humidity, and bulb status.
  - Enabled automatic and remote control of fan and bulb, improving energy efficiency and user convenience.
  - Integrated sensors and microcontroller to reduce manual intervention and enhance system reliability.

- **Regenerative Braking System**
  - Designed and implemented a regenerative braking system to capture and store kinetic energy during braking, improving overall energy efficiency.
  - Used electrical and electronic components to convert mechanical energy into usable electrical energy.
  - Tested the system to ensure reliable braking performance and effective energy recovery.

- **Li-Fi Technology**
  - Designed and implemented a Li-Fi communication system using visible light for high-speed data transmission.
  - Developed a secure and efficient alternative to traditional wireless communication.
  - Enhanced data transfer speed, reliability, and overall system performance

- **Food Cafe Application**
  - Designed and developed a Java-based Food Cafe Management System to streamline cafe operations and order handling.
  - Implemented features for menu management, customer order processing, billing, and payment handling.
  - Improved operational efficiency and customer service through a user-friendly interface and structured reporting system.